

Literature review and preliminary actuator cylinder modelling of adjacent vertical-axis wind turbines

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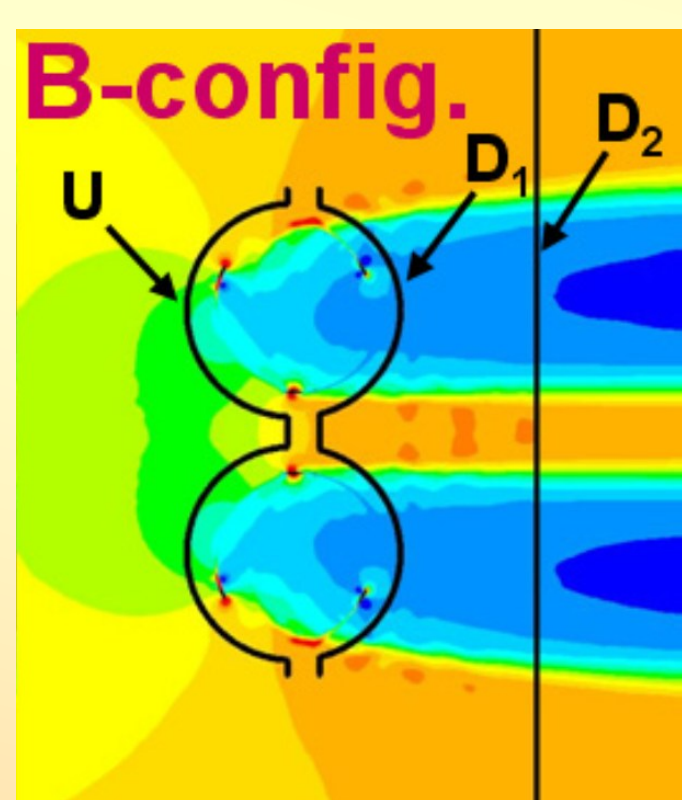
Abstract

Vertical-axis wind turbines (VAWTs) can increase in power output when placed in pairs or rows, both due to the blockage effect and VAWT-specific features. The physics of this power gain, some data on how much power can be expected, and some of the other consequences on the flow are all discussed, as functions of various turbine/pair parameters. Additionally, the Actuator Cylinder model is discussed. This extension of the actuator disk method is low-complexity but medium-fidelity, using turbine forces to predict the flow field and power generated. My own preliminary code is discussed and shared.

Power Gain Mechanisms [FDM]

On the upwind half of the blade revolution:

- the flow is impeded, reducing flow rate and power...
- ...but the flow direction changes, changing the angle of attack, benefitting power generation

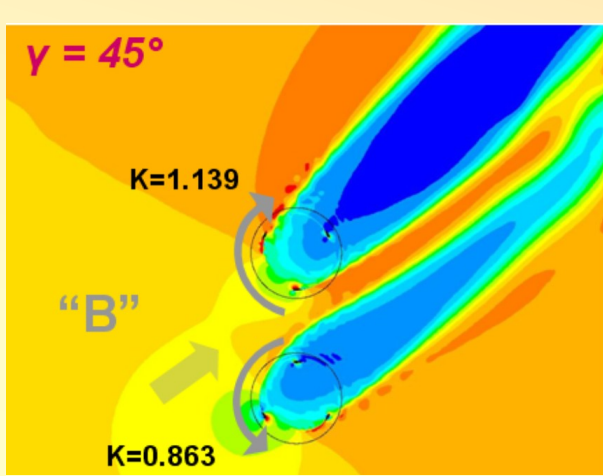


The latter effect dominates, and there is a **net increase in power**.

On the downwind half of the blade revolution, the flow cannot freely 'escape' laterally, increasing flow rate, **increasing power generated**.

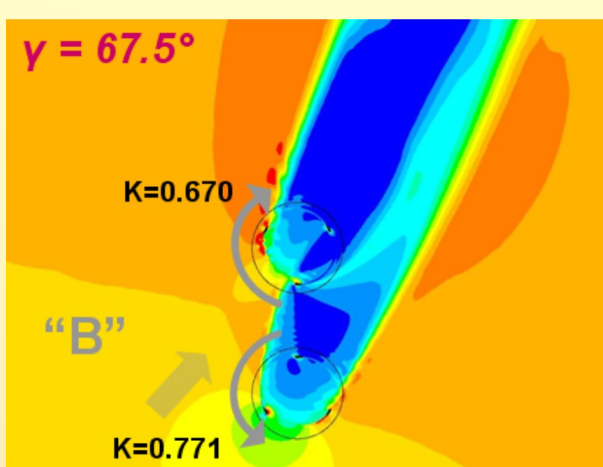
At small yaw angles, the **power increases** overall, as:

- The front turbine's wake is obstructed by the back turbine, decreasing its power
- The back turbine is hit by accelerated flow from around the front turbine, increasing its power more than the front turbine loses power



At large yaw angles, the power of both turbines **decreases**:

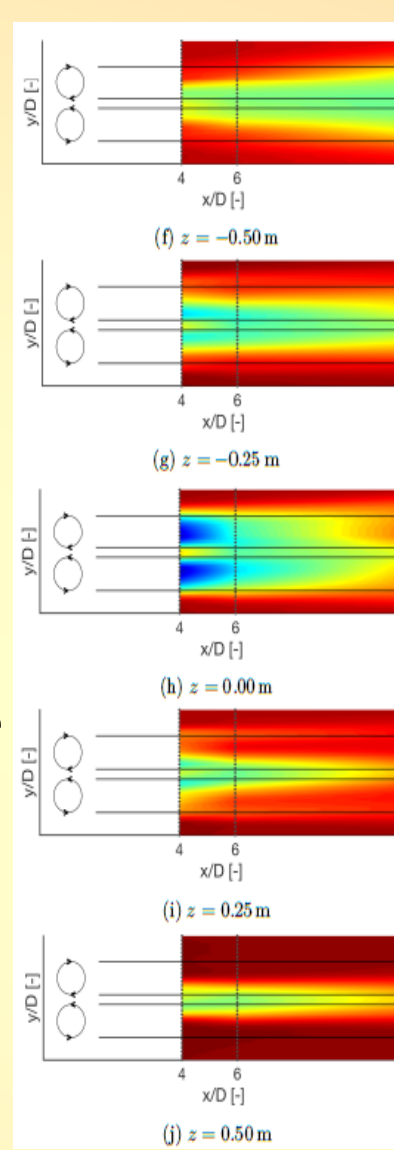
- The front turbine's wake is further obstructed, further decreasing power
- The back turbine becomes shadowed by the front turbine, decreasing its power instead of increasing it.



Other Interesting Physics

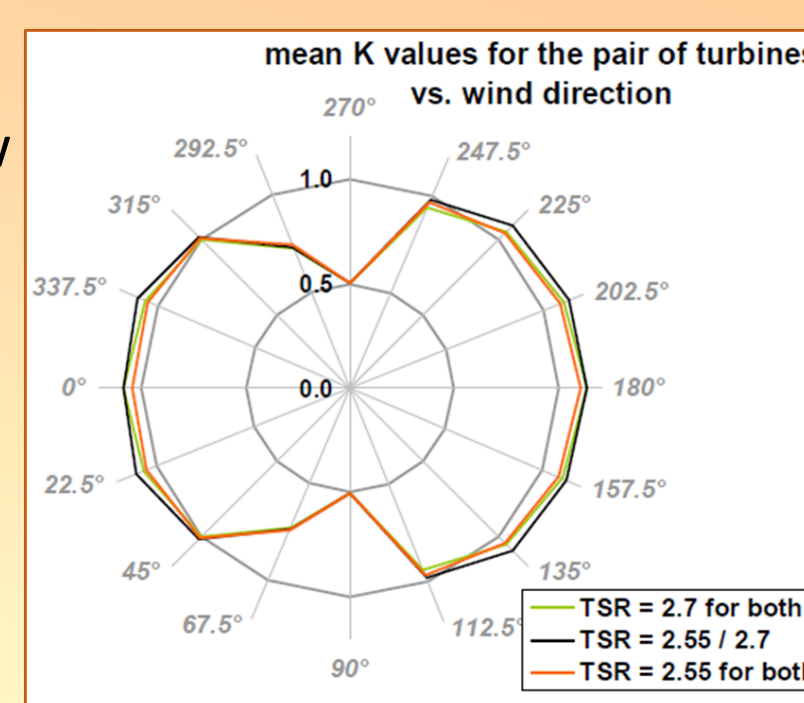
Adjacent VAWTs can be arranged in one of three ways: clockwise-anticlockwise, AC-C, or co-rotating.

- Isolated VAWTs' wakes bend. In pairs, depending on rotation direction, they may diverge, resulting in a weaker, wider wake; or converge, resulting in a stronger, narrower wake with vertical mixing.
- Upwind-travelling and downwind-travelling blades see different effective wind speeds, so distorted flow will have a different interaction with each.



These factors affect wake structure, power and yaw sensitivity. [Wake]

When **tip-speed ratio** (TSR) increases, the flow concentration benefit decreases, but the AoA benefit increases. For a small increase in TSR past the isolated turbine's optimum, the benefit dominates; hence, the optimum TSR of VAWTs in a pair is higher than the same VAWT in isolation.

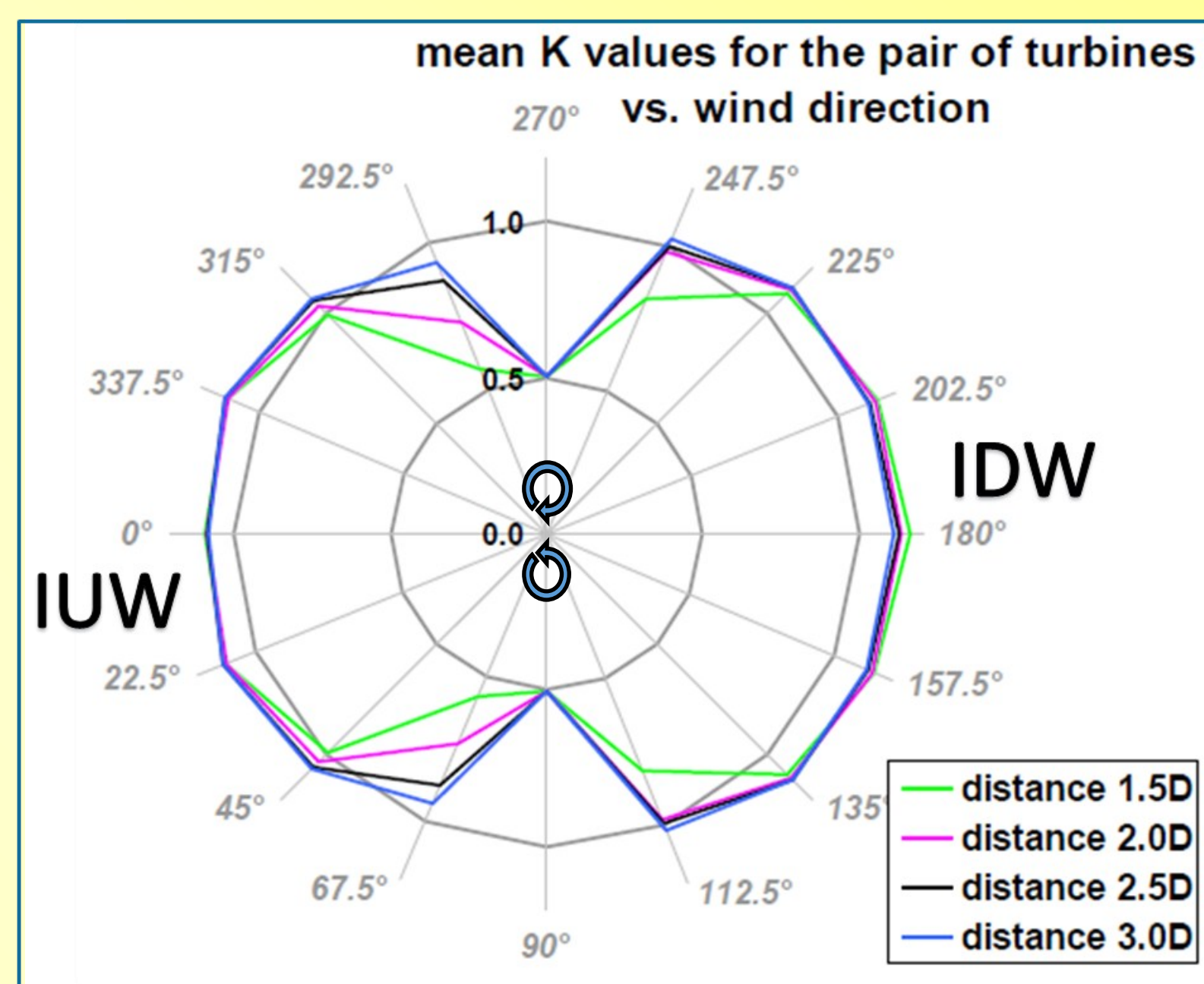


In yawed wind, only the back turbine acts like it's in a pair; hence, controlling TSR as a function of yaw angle can further increase power in a VAWT pair. [FDM]

Power Increase: Literature

Shown in the figure to the side are CFD results from [FDM], showing power of a pair of turbines in various wind directions with various separations between them. These results are discussed, alongside experimental results from [ExpVal] and CFD results from [ScaleUp] (see Sources below).

Here, IUW ("inner-upwind") refers to the blades travelling upwind on the inside of the VAWT pair, and similar for IDW ("inner-downwind"); and "1.5D", "2.0D" etc. refers to the distance between the VAWT axes in units of turbine diameters (with 1D being the tightest possible spacing).



In direct wind, power increases by around 16%. This is in line with work by [ExpVal], where power increases by 13.1—15.9% for tighter (1.2D and 1.3D) separations, and higher than [ScaleUp]'s work finding increases of ~10% in direct wind.

The tightest separation produces the greatest power in direct wind, but is the most yaw-sensitive.

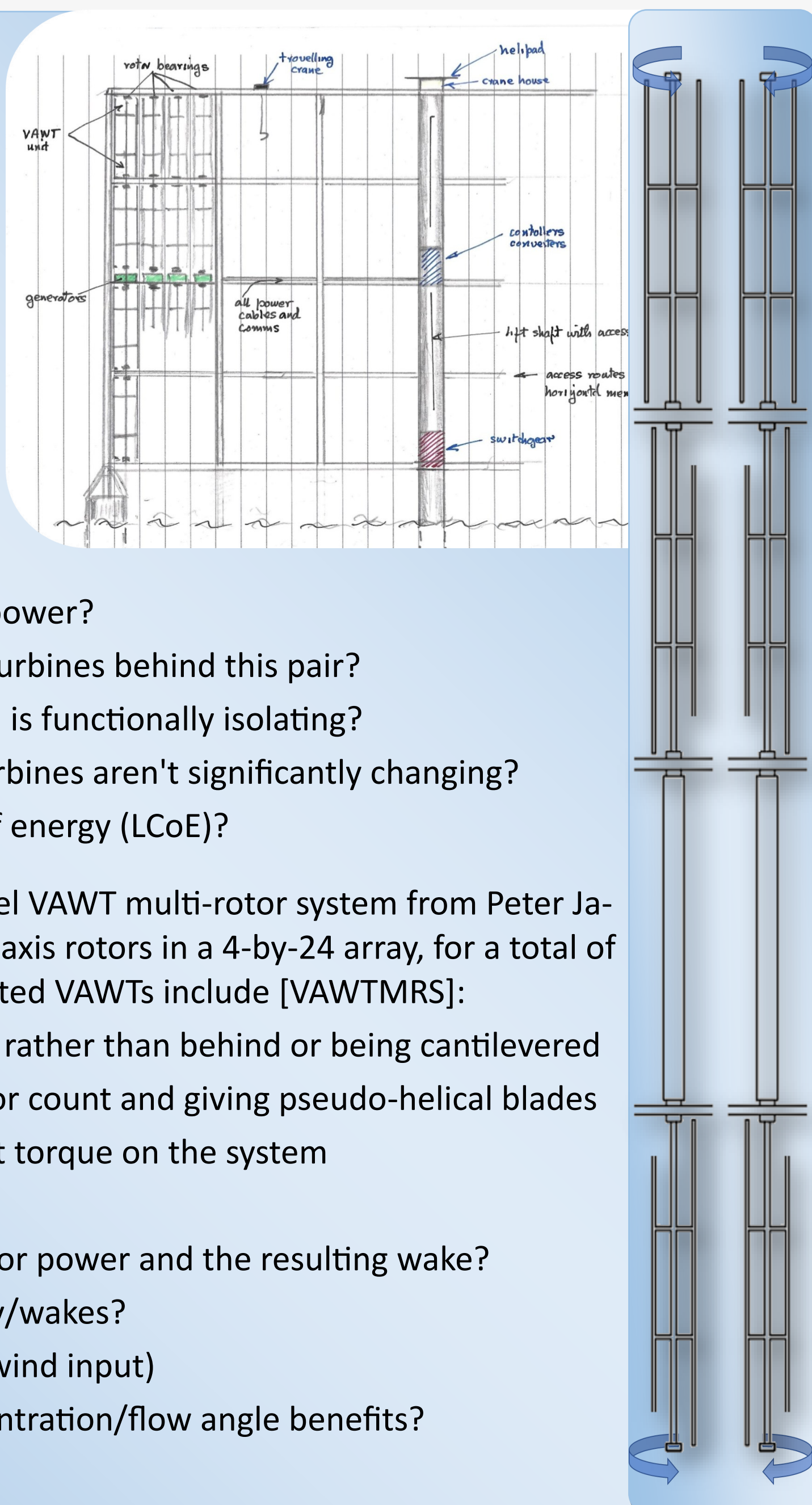
Notable is the impressive yaw insensitivity, with similar power gains over a wide range of angles, and at least some power gain over an even wider range. This agrees with [ScaleUp], where power increased for almost all configurations they considered, except for direct shadowing, and the tightest separation (1.375D) at the steepest angle (60°).

Though this plot shows 0° IDW as producing the greatest power, [ScaleUp] usually found a greater power gain with an oblique wind angle, as does the IUW case here.

Future Work and Application

My own research hopes to improve my actuator model, up to and hopefully past what exists in the literature already...

- introduce nonlinearities
- introduce a better wake modelling method
- consider adding virtual camber
- extend to 3D
- include multiple rows + stacked turbines



...to obtain results of general interest...

- What rotation direction setup is optimal for power?
- How do different rotation directions impact turbines behind this pair?
- What separation is optimal? What separation is functionally isolating?
- How many in a row before the end/centre turbines aren't significantly changing?
- How do these features affect levelized cost of energy (LCoE)?

...and results specifically for the context of a novel VAWT multi-rotor system from Peter Jamieson. This system features 96 500kW vertical-axis rotors in a 4-by-24 array, for a total of 48MW. Advantages over HAWT MRSes and isolated VAWTs include [VAWTMRS]:

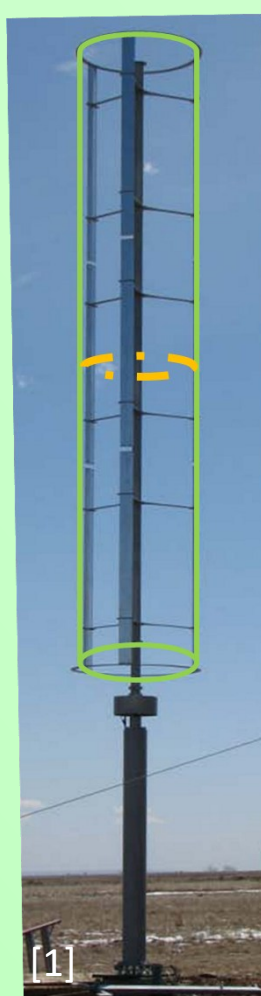
- rotors being attached from above and below, rather than behind or being cantilevered
- vertically-connected rotors reducing generator count and giving pseudo-helical blades
- Counter-rotating rotors cancelling out the net torque on the system

Research questions include:

- What rotation direction combination is best for power and the resulting wake?
- How does tight spacings affect yaw sensitivity/wakes?
- What are 3D effects? (Vertical flow, sheared wind input)
- How do structural supports affect flow concentration/flow angle benefits?

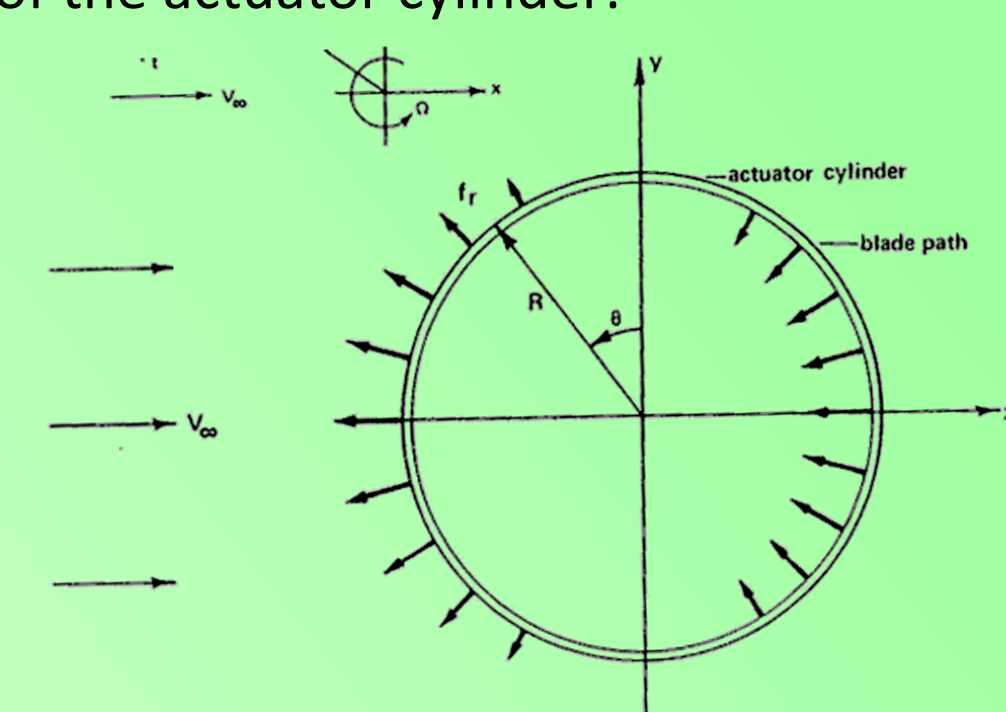
The Actuator Cylinder Method

Elementary analysis of a horizontal axis turbine is often done using an actuator disk, replacing a turbine rotor by an idealised vertical circle that extracts energy. This derives, for example, the Betz limit, $C_{p,max} = 16/27 \approx 59.3\%$.



Simplified analysis of a VAWT can be done using a similar concept, that of the actuator cylinder:

- approximate a VAWT cross-section by a horizontal circle which extracts energy as wind flows through its front and back
- use mass and momentum conservation equations to determine formulae for the wind pressure and wind speed
- find wind pressure and speed based on turbine forces



This code is simple to construct, and gives high accuracy relative to the computational time.

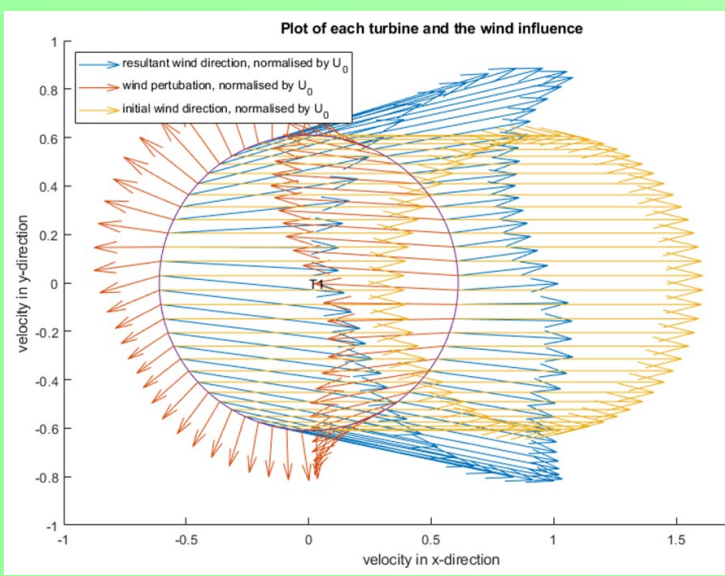
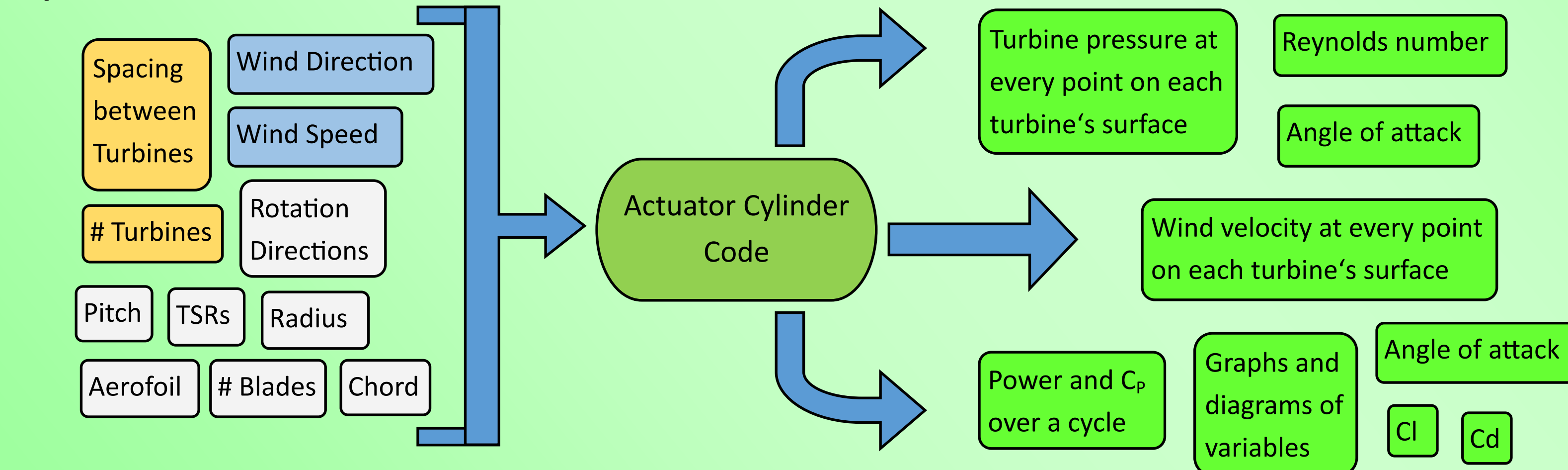
The most basic version of the code neglects nonlinear terms, and can calculate the velocity at any point using a prescribed turbine pressure $q(\theta)$. [OriginalAC]

Typical modern actuator codes also: introduce a nonlinear correction term, and use aerofoil data to calculate $q(\theta)$ at a given point. The former increases accuracy at high loadings, with less complexity than a full nonlinear approach. The latter requires an iterative loop to find $v = f(q(v))$ across the cylinder, and switches from an integral of $q(\theta)$ to a summation across some number of $q(\theta)$.

More advanced modifications include:

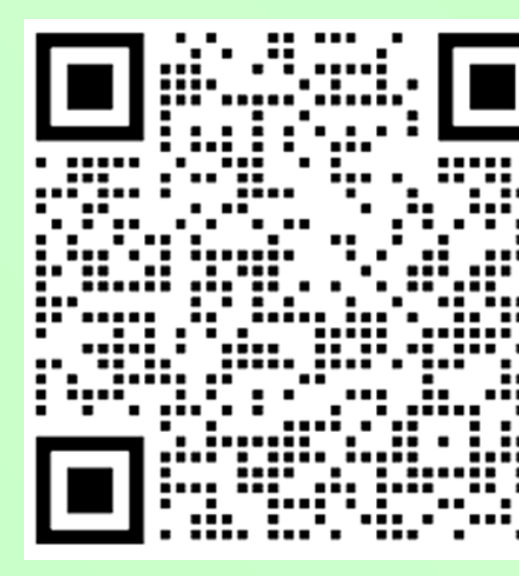
- Full nonlinear: large increase in computational resources, requiring resolving the full flow field and an iterative loop.
- Combination of other codes: [ACFOAM] uses an Actuator Cylinder code to determine turbine forces, and feeds them into OpenFOAM to resolve the wake field.
- Other corrections: eg introducing virtual camber into the $q(\theta)$ derivation, or using a different wake model.

My code



For my PhD, I've constructed an Actuator Cylinder code in Matlab, inputting the above variables and outputting the above information. It can be run in bulk, comparing outputs of different runs.

Though it's a preliminary code (see "Future Work and Application"), results are qualitatively correct (see left) and it's currently being updated. It's available via the QR code (right), or at github.com/AngusBuck/Multiple-Actuator-Cylinder; each new update will be shared.



Sources

[FDM]: Stefania Zanforlin, Takafumi Nishino, Fluid dynamic mechanisms of enhanced power generation by closely spaced vertical axis wind turbines, Renewable Energy, Volume 99, 2016, Pages 1213-1226, ISSN 0960-1481, <https://doi.org/10.1016/j.renene.2016.08.015>. (<https://www.sciencedirect.com/science/article/pii/S0960148116307169>)

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[MultiAC]: Ning, A. (2016). Actuator cylinder theory for multiple vertical axis wind turbines. Wind Energy Science, 1(2), 327–340. <https://doi.org/10.5194/wes-1-327-2016>

[ExpVal]: Vergaerde, A., De Troyer, T., Standaert, L., Kluczevska-Bordier, J., Pitance, D., Immas, A., Silvert, F., Runacres, M. (2020). Experimental validation of the power enhancement of a pair of vertical-axis wind turbines. Renewable Energy, 146, 181-187. [10.1016/j.renene.2019.06.115]. <https://doi.org/10.1016/j.renene.2019.06.115>

[Wakes]: Antoine Vergaerde et al 2020, Influence of the direction of rotation on the wake characteristics of closely spaced counter-rotating vertical-axis wind turbines, J. Phys.: Conf. Ser. 1618 062017

[ScaleUp]: Joachim Toffegaard Hansen, Mahak Mahak, Iakovos Tzanakis, Numerical modelling and optimization of vertical axis wind turbine pairs: A scale up approach, Renewable Energy, Volume 171, 2021, Pages 1371-1381, ISSN 0960-1481, <https://doi.org/10.1016/j.renene.2021.03.001>. (<https://www.sciencedirect.com/science/article/pii/S096014812100344X>)

[ACFOAM]: Martinez, Edgar & Ordaz, Francisco & Sen, Mihir. (2021). Vertical-axis wind-turbine computations using a 2D hybrid wake actuator-cylinder model. 10.5194/wes-2021-38.

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[4] Bremseth, J., Duraisamy, K. Computational analysis of vertical axis wind turbine arrays. Theor. Comput. Fluid Dyn. 30, 387–401 (2016). <https://doi.org/10.1007/s00162-016-0384-y>

[5] Takafumi Nishino and Scott Draper 2015, Local blockage effect for wind turbines, J. Phys.: Conf. Ser. 625 012010